

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461167.7038 +/- 0.0071 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.053
Transit depth (Rp/Rs)^2	3.21 +/- 1.9 [%]
Orbital Inclination (inc)	87.44 +/- 1.77
Airmass coefficient 1 (a1)	1.36 +/- 0.82
Airmass coefficient 2 (a2)	-0.32 +/- 0.2
Scatter in the residuals of the lightcurve fit is	58.0 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.5
Transit Duration (day)	0.096 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.053 vs 0.1326 (deviation 0.65σ)
Duration fit vs published	0.096 d vs 0.1238 d (deviation 0.86σ)
Mid-time O–C	-5.0 min
Depth SNR	2.5
Residual scatter	58.0 %

Light curve

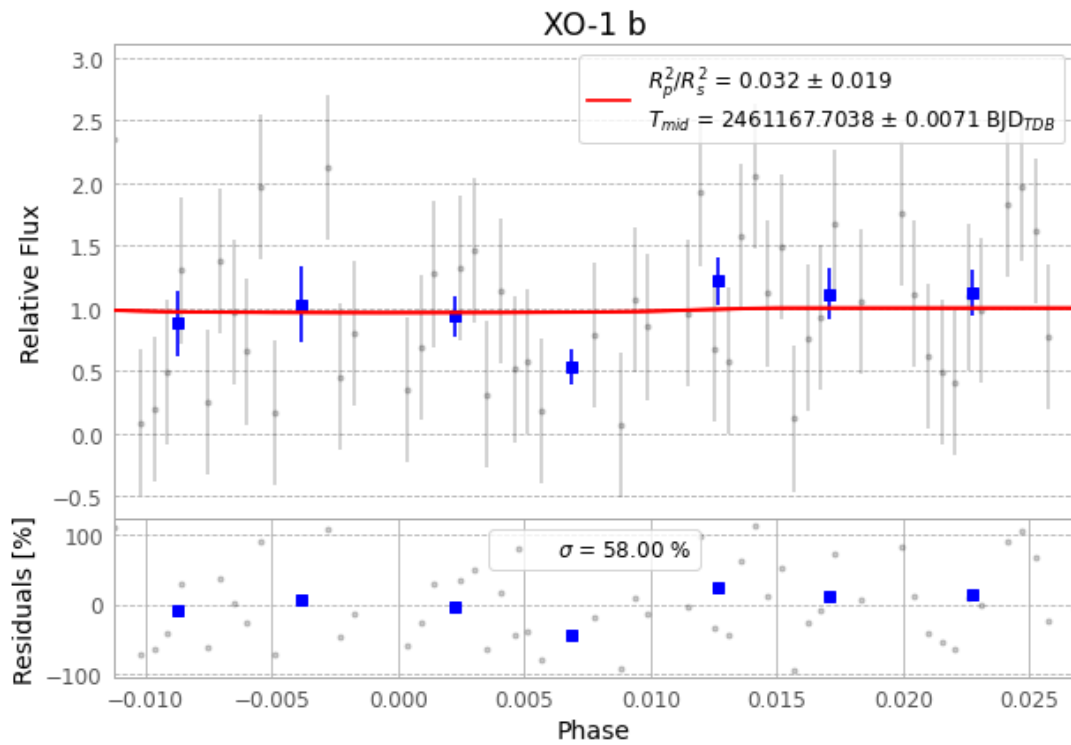


Figure 1 — FinalLightCurve_XO-1 b_2026-05-06.png (SHA-256 3f12887741001c18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 245], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8f3e37da5498ac22...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

78 FITS frames (51 MB), XO-1260507032715.FITS → XO-1260507071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-260507.log (6126e56519b1...), FinalLightCurve_XO-1 b_2026-05-06.png (3f1288774100...), FinalLightCurve_XO-1 b_2026-05-06.csv (5af0a11b3bbd...)

Compute resources

Wall time 145.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.